

Library Management System

Project Title:

Library Management System

Objective

To develop a console-based Library Management System that allows a librarian to manage books by adding, viewing, issuing, returning, searching, and counting available books.

Tools and Technologies

- Python 3.11.9
- Visual Studio Code

Project Structure

LibraryManagement/

- main.py
- book.py
- library_utils.py

Modules Description

book.py

Contains the Book class with the following attributes:

- book_id
- title
- author
- is_issued

Methods:

- issue_book()
- return_book()
- display()

Code:

```
class Book:
    def __init__(self, book_id, title, author):
        self.book_id = book_id
        self.title = title
        self.author = author
        self.is_issued = False

    def issue_book(self):
        if self.is_issued:
            print("Book is already issued.")
        else:
            self.is_issued = True
            print("Book issued successfully.")
```

```
    def return_book(self):
        if not self.is_issued:
```

```
        print("Book was not issued.")
    else:
        self.is_issued = False
        print("Book returned successfully.")
```

```
def display(self):
    status = "Issued" if self.is_issued else "Available"
    print(f"ID: {self.book_id}")
    print(f"Title: {self.title}")
    print(f"Author: {self.author}")
    print(f"Status: {status}")
    print("-----")
```

library_utils.py

Contains utility functions:

- search_book()
- count_available_books()
- display_all_books()

Code:

```
def search_book(book_list, book_id):
    for book in book_list:
        if book.book_id == book_id:
            return book
    return None
```

```
def count_available_books(book_list):
    count = 0
    for book in book_list:
        if not book.is_issued:
            count += 1
    return count
```

```
def display_all_books(book_list):
    if len(book_list) == 0:
        print("No books available in library.")
    return
```

```
print("\n==== ALL BOOKS =====")
for book in book_list:
    book.display()
```

main.py

Provides a menu-driven interface to:

- Add Book
- View All Books
- Issue Book
- Return Book
- Search Book
- Count Available Books
- Exit

Features

1. Add new books to the library.
2. Display all available books.
3. Issue books to users.
4. Return issued books.
5. Search books using Book ID.
6. Count available books.
7. User-friendly menu-driven interface.

Code:

```
from book import Book
from library_utils import (
    search_book,
    count_available_books,
    display_all_books
)

book_list = []

while True:

    print("\n===== LIBRARY MANAGEMENT SYSTEM =====")
    print("1. Add Book")
    print("2. View All Books")
    print("3. Issue Book")
    print("4. Return Book")
    print("5. Search Book")
    print("6. Count Available Books")
    print("7. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":

        book_id = int(input("Enter Book ID: "))
        title = input("Enter Book Title: ")
        author = input("Enter Author Name: ")

        new_book = Book(book_id, title, author)
        book_list.append(new_book)

        print("Book added successfully!")

    elif choice == "2":

        display_all_books(book_list)

    elif choice == "3":

        book_id = int(input("Enter Book ID to issue: "))

        book = search_book(book_list, book_id)

        if book:
            book.issue_book()
        else:
            print("Book not found.")

    elif choice == "4":

        book_id = int(input("Enter Book ID to return: "))

        book = search_book(book_list, book_id)
```

```
    if book:
        book.return_book()
    else:
        print("Book not found.")
```

```
elif choice == "5":
```

```
    book_id = int(input("Enter Book ID to search: "))
```

```
    book = search_book(book_list, book_id)
```

```
    if book:
        print("\nBook Found")
        print("-----")
        book.display()
    else:
        print("Book not found.")
```

```
elif choice == "6":
```

```
    available_books = count_available_books(book_list)
```

```
    print("Available Books:", available_books)
```

```
elif choice == "7":
```

```
    print("Thank you for using Library Management System.")
    break
```

```
else:
```

```
    print("Invalid choice! Please try again.")
```

Output

The system successfully manages library records and performs book operations efficiently through a console-based application.

1. Add new books to the library.

```
• (.venv) PS C:\Users\91831\OneDrive\Documents> cd LibraryManagement
○ (.venv) PS C:\Users\91831\OneDrive\Documents\LibraryManagement> python main.py

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 1
Enter Book ID: 101
Enter Book Title: ai
Enter Author Name: Nandita
Book added successfully!

...

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 1
Enter Book ID: 102
Enter Book Title: python
Enter Author Name: Naveen
Book added successfully!
```

2. Display all available books.

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 2

===== ALL BOOKS =====
ID: 101
Title: ai
Author: Nandita
Status: Available
-----
ID: 102
Title: python
Author: Naveen
Status: Available
-----
ID: 103
Title: java
Author: anil
Status: Available
-----
```

3. Issue books to users. and Return issued books.

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 3
Enter Book ID to issue: 102
Book issued successfully.

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 4
Enter Book ID to return: 102
Book returned successfully.
```

4. Search books using Book ID.

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 5
Enter Book ID to search: 102

Book Found
-----
ID: 102
Title: python
Author: Naveen
Status: Available
-----

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 5
Enter Book ID to search: 107
Book not found.
```

6. Count available books.

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 6
Available Books: 3

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. View All Books
3. Issue Book
4. Return Book
5. Search Book
6. Count Available Books
7. Exit
Enter your choice: 7
Thank you for using Library Management System.
(.venv) PS C:\Users\91831\OneDrive\Documents\LibraryManagement>
```

Conclusion

The Library Management System was successfully developed using Python. The project demonstrates the use of classes, objects, functions, modules, conditional statements, loops, and file organization in Python. It provides a simple and effective solution for managing library books.